

Weekly Progress Report

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Domain: Artificial Intelligence And Machine Learning

Date of Submission: 08/07/2026

Week Ending: 02

I. Overview:

This week focused on model evaluation, Streamlit application development, project optimization, GitHub integration, and cloud deployment.

II. Achievements:

1. Model Evaluation

- Evaluated all trained machine learning models.
- Generated confusion matrices.
- Created model accuracy comparison chart.
- Selected the best performing model.

2. Prediction System

- Developed gearbox health prediction module.
- Implemented confidence score generation.
- Displayed extracted statistical features after prediction.

3. Streamlit Web Application

Developed a complete interactive dashboard with:

- File upload functionality
- Health prediction
- Confidence percentage
- Feature visualization

- Accuracy comparison graph
- User-friendly interface

4. Project Organization

- Restructured project folders.
- Improved code modularity.
- Added requirements.txt.
- Updated README documentation.

5. GitHub & Deployment

- Uploaded complete project to GitHub.
- Created professional repository.
- Deployed application on Streamlit Cloud.
- Successfully tested live deployment.

III. Challenges:

1. Streamlit Integration

- Resolved file path issues.
- Fixed prediction function errors.
- Improved UI layout.

2. Deployment

- Solved dependency issues.
- Configured project for cloud deployment.

IV. Learning Resources

- Streamlit Official Documentation
- GitHub Documentation
- Scikit-Learn Documentation
- Python Official Documentation

VI. Additional Comments:

“Successfully completed the complete end-to-end Machine Learning pipeline including preprocessing, feature extraction, model training, evaluation, web application development, GitHub integration, and cloud deployment. The project is fully functional and deployment-ready.”